

Evaluation report

How I measured the result

I compared predictions with H1 labels by invoice ID. All 10 records sharing five duplicate IDs were excluded, leaving **908 eligible invoices**. H1 labels and previous mistakes informed development: these results are not an independent test. Reference groups prevent using an invoice's own price to infer its service; they do not undo that development exposure.

The first full-H1 baseline answered **40/908 (4.4%): 37 correct, 3 incorrect**, with 868 unanswered. The final results below include rule improvements and validated, cached AI evidence; they are not a JSON-only comparison.

Error/no-error decisions: 908/908 correct, including all 53 erroneous invoices. **Corrected totals: 876 supplied and correct; 32 unanswered**. Error-category sets matched exactly on 885/908 invoices, but only **30/53 erroneous invoices**. Getting the verdict right is therefore not the same as completing the audit.

Across all hospitals, **3,651 of 4,824 eligible invoices were answered**; 1,173 remain unanswered, plus 62 duplicate-ID records excluded from submission. The H2-H5 submission contains **2,743 opinions out of 3,916 eligible invoices**. H2-H5 have no labels, so these are coverage figures, not measured accuracy.

Category results, simplified

12 of 17 error categories had no missed or extra tags. These cover arithmetic and totals, contract numbers, duplicate charges, malformed/out-of-window dates, bundles, daily caps, premiums and volume discounts. The five needing work are:

1. **Wrong price:** found 10/10 labelled cases, but added 17 extra tags.
2. **Unknown service:** found 8/12 cases; missed 4, with no extra tags.
3. **Wrong billing unit:** found 10/10 cases, with 1 extra tag.
4. **Service after invoice date:** found 5/5 cases, with 2 extra tags.
5. **Exclusion window:** found 0/3 cases; the boundary interpretation remained unresolved.

These count error tags, not distinct invoices; several tags can apply to one invoice. An extra tag is a mismatch with the labels, not necessarily a wrong error/no-error decision. Full counts, precision and recall for every original category remain in the evaluation JSON referenced below.

Four ways the approach still falls short

1. **The right verdict, but an over-broad explanation.** INV-H1-000186 correctly identifies an unearned discount and arithmetic error, but also adds a generic wrong-price tag. Its corrected total is right. This pattern explains all 17 extra wrong-price tags. Next: define when a specific explanation should replace a generic one, without hiding separately supported errors.
2. **Missing words can lead to the wrong service match.** INV-H1-000236 says "Fract Outpatient Radiotherapy"; the checker assumes the missing modifier "Metabolic". The label instead treats it as an unknown service. A short list of possible matches is not proof. Next: require stronger evidence for clinical modifiers or request a provider service-code crosswalk.
3. **A clause can leave a boundary unclear.** INV-H1-000211 has relevant services exactly 10 days apart. The contract says "within" 10 days without explicitly defining that endpoint. Labels include it; the checker withholds that category. All three missed exclusions have this pattern. Next: obtain or clearly declare a boundary interpretation, rather than present it as a contract fact.
4. **Finding an error does not reveal the correct amount.** INV-H1-000049 bills 15 hours against a 12-hour cap. The cap proves excess, but not the intended quantity; simply changing 15 to 12 does not recover the labelled total. Next: seek the underlying service record. All 32 withheld totals belong to erroneous invoices; none of the supplied H1 totals is wrong.

Confidence and next steps

Confidence reflects evidence strength, **not a calibrated probability**. Weaker matches, assumptions and missing totals lower it. The development-only confidence diagnostic is retained in the detailed results; it does not establish calibration on unseen hospitals. Date quarantine isolates malformed dates from dated history checks, not from every possible effect; uncertainty remains where it matters.

With another week, I would prioritize authoritative unit and date-boundary clarifications, stronger service matching, and independently reviewed examples for testing and confidence calibration. I would not force unresolved invoices into 0/1 answers merely to increase coverage.

Evidence and reproduction

Run `python3 main.py` for a cached-only audit and the six-column submission. Full per-category results for this snapshot: `audit_output/test_all_20260905_submission_final/evaluation.json`. New runs also produce `evaluation.json`. See `docs/DECISION_LOG.md`, `prompts/README.md` and `AI_USAGE.md` for assumptions and development provenance.